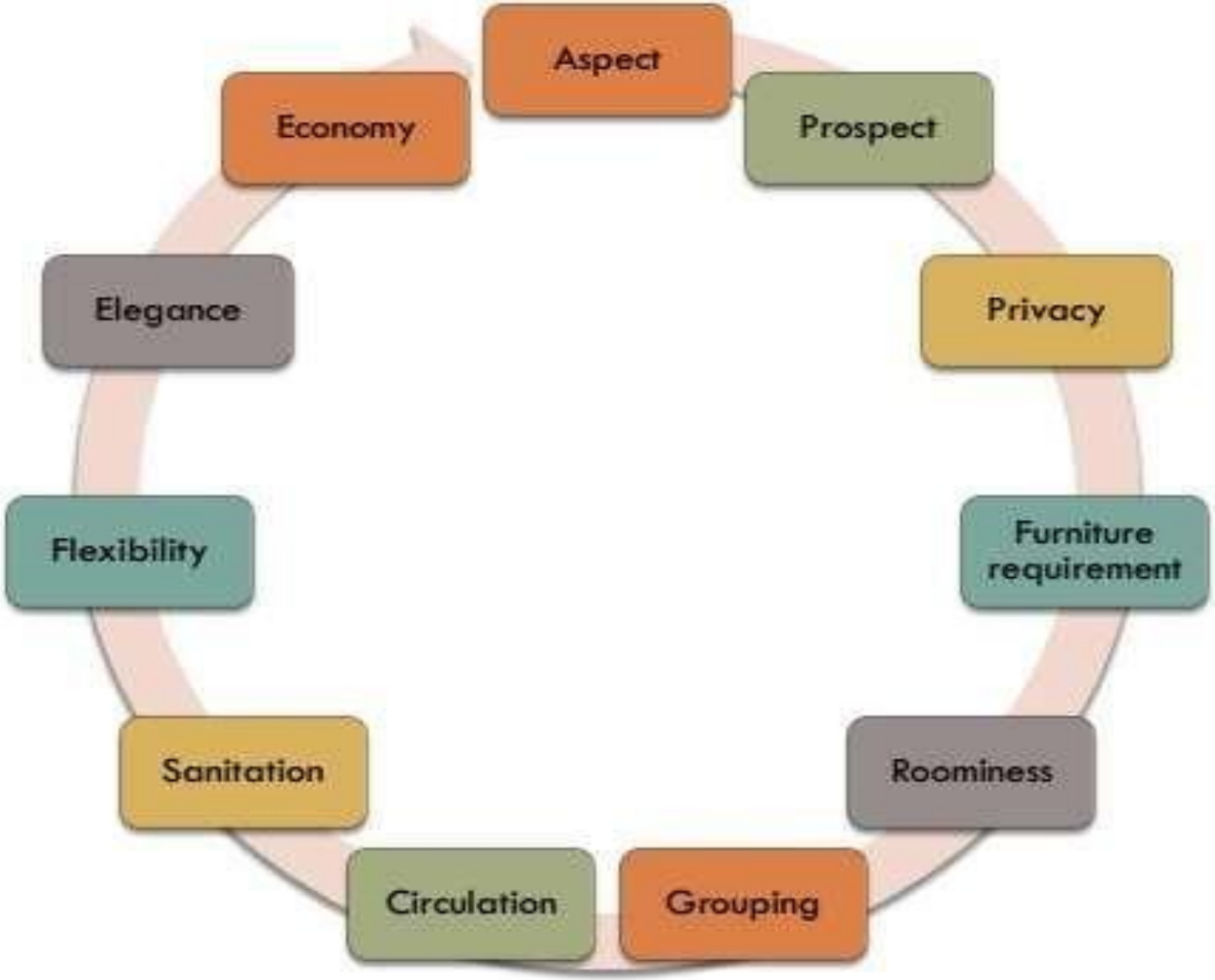
Planning of building depends on:

- Human habitation & their requirements Component parts, sizes and inter- relationship Topography and shape of plot, Climatic condition, Location and neighborhood.
- GROUPING and ARRANGEMENT of components of building in a systematic manner so as to form a homogeneous body with a comprehensive look out to meet its functional purpose.

1.1 Principles of Building Planning

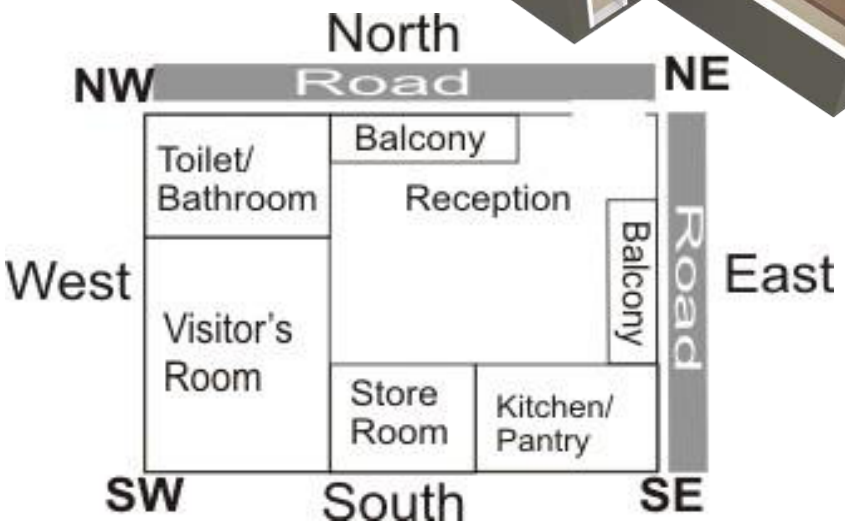


Principles of Building Planning[Continued…]

ASPECT- Placement of different rooms of house in accordance with our activities at different hours of day.

Rooms should get enough sunlight and air, which gives Cheerful atmosphere, Comfort, Hygienic condition.





PROSPECT- is the view from outside of a house. The house should have a proper prospect so that it can give a feeling of cheerfulness to the people living in it. It should create a good impression on a person who views it from outside.





Principles of Building Planning[Continued…]

PRIVACY- Privacy can be of Sight (bath, w/c) Sound (confidential discussion, study room) Both sight and sound (bed room)

Divides as-

- Internal Privacy (Inside the plan)
- External Privacy (Outside the plan)





FURNITURE REQUIREMENT : Planner should know how much space is needed by each function Room sizes can be completed on basis of permanent furniture to be used, It's dimensions and arrangement Clearance for circulation.



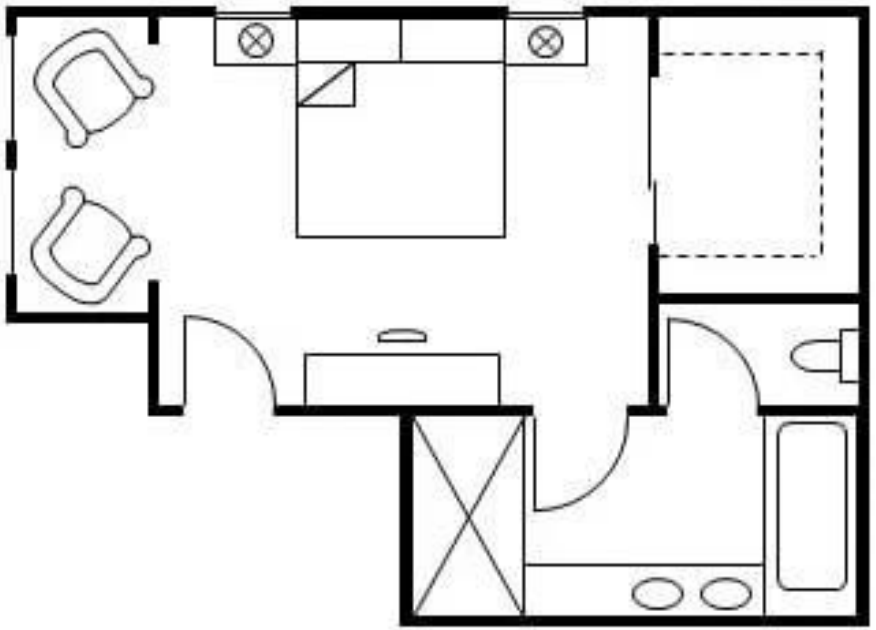


Source- Shutterstock

Principles of Building Planning[Continued…]

ROOMINESS- It is feeling created after a room is well-furnished with all permanent furniture as spacious and well-planned.

- Max use of a room with min possible dimensions.
- Rectangular room gives better outlook compare to square room of same floor area.
- Length/breadth ratio of 1.2 to 1.5 is desirable



GROUPING- : It is arrangement of different rooms with reference to their functions, it improves comfort, privacy and convenience and minimizes circulation.

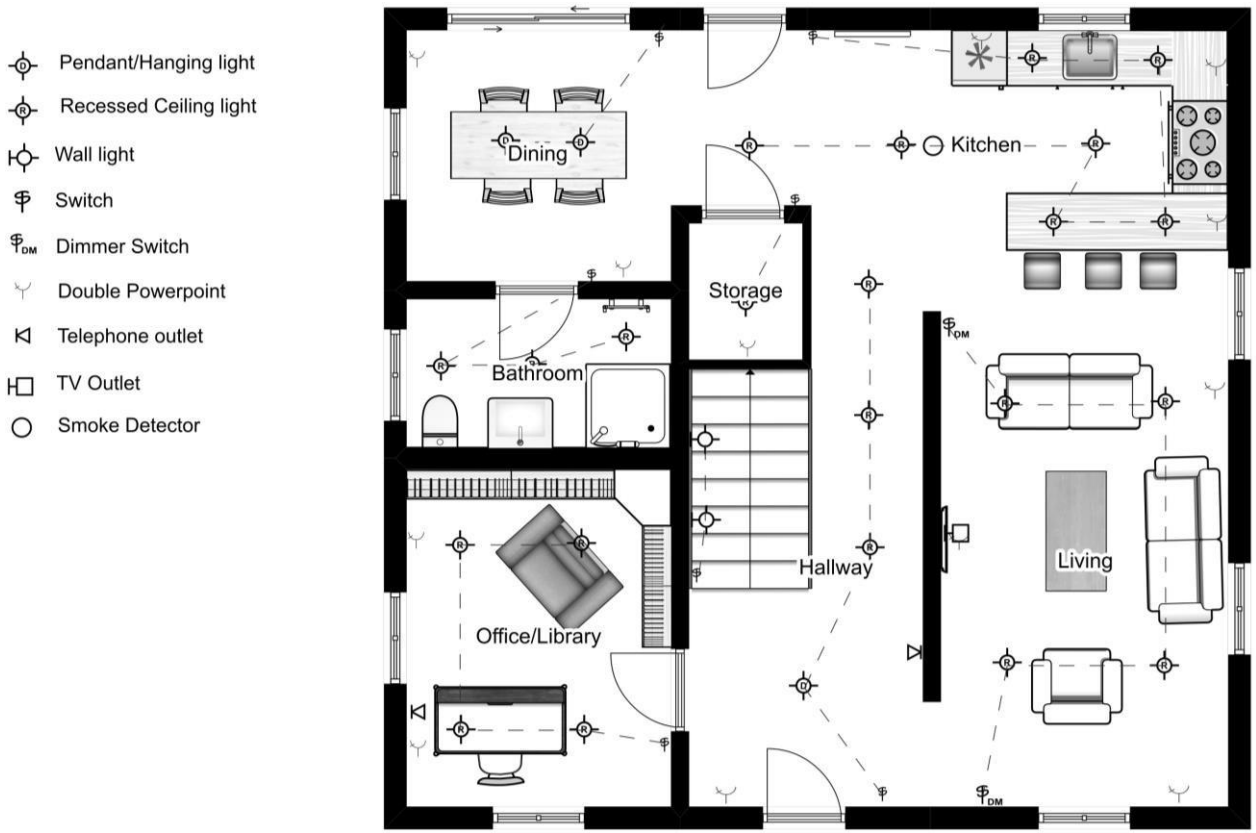


Source- Pinterest

Principles of Building Planning[Continued…]

CIRCULATION- It is access into or out of a room, it is internal movement inside a building.

Circulation area shall be straight, short, bright, lighted both day and night, well ventilated and free from obstructions.

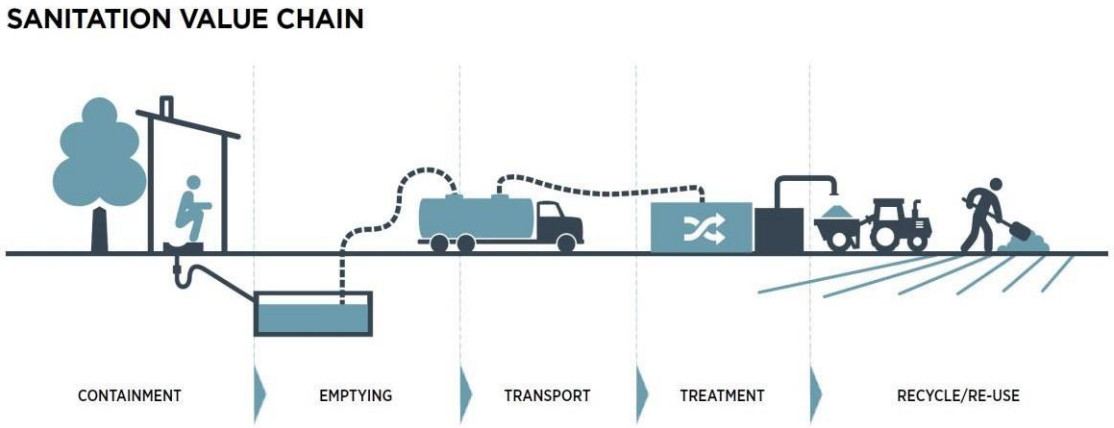


SANITATION : It is provision and upkeep of various components of house to keep inmates cheerful and free from disease.

Factors influence sanitation are-

- Lighting
- Ventilation
- Cleanliness





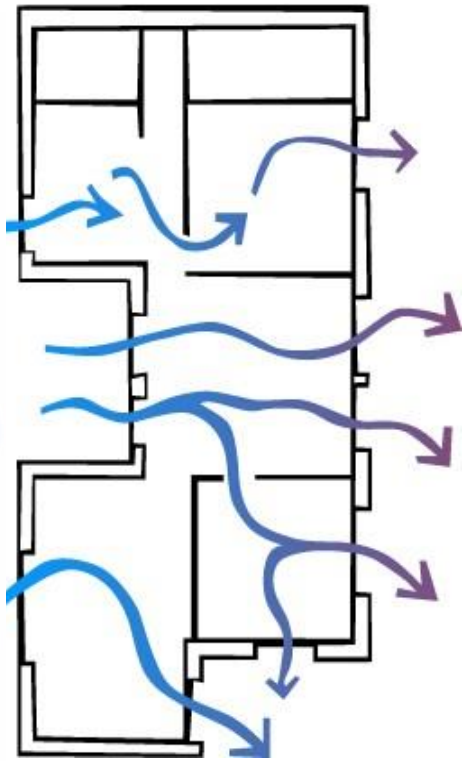
Source- Pinterest

Principles of Building Planning[Continued…]

VENTILATION- A system of supplying or removing air by natural or mechanical means to or from any enclosed space to create and maintain comfortable condition.

Orientation of building and location of windows help in providing proper ventilation.





FLEXIBILITY- A room which is planned for one function be used for other, if sorequired.

It is ease with which a room designated for a particular activity can accommodate more load temporarily or may supplement activity of another room.



Source- SustainableStyles

Principles of Building Planning

ELEGANCE- It is grand appearance of a building, mainly owing to the elevation which in turn depends on plan.

A better elegance can be obtained by-

Superior building materials for facing – like paint, glass, timber, polished stones, granite, marble, mosaic etc.



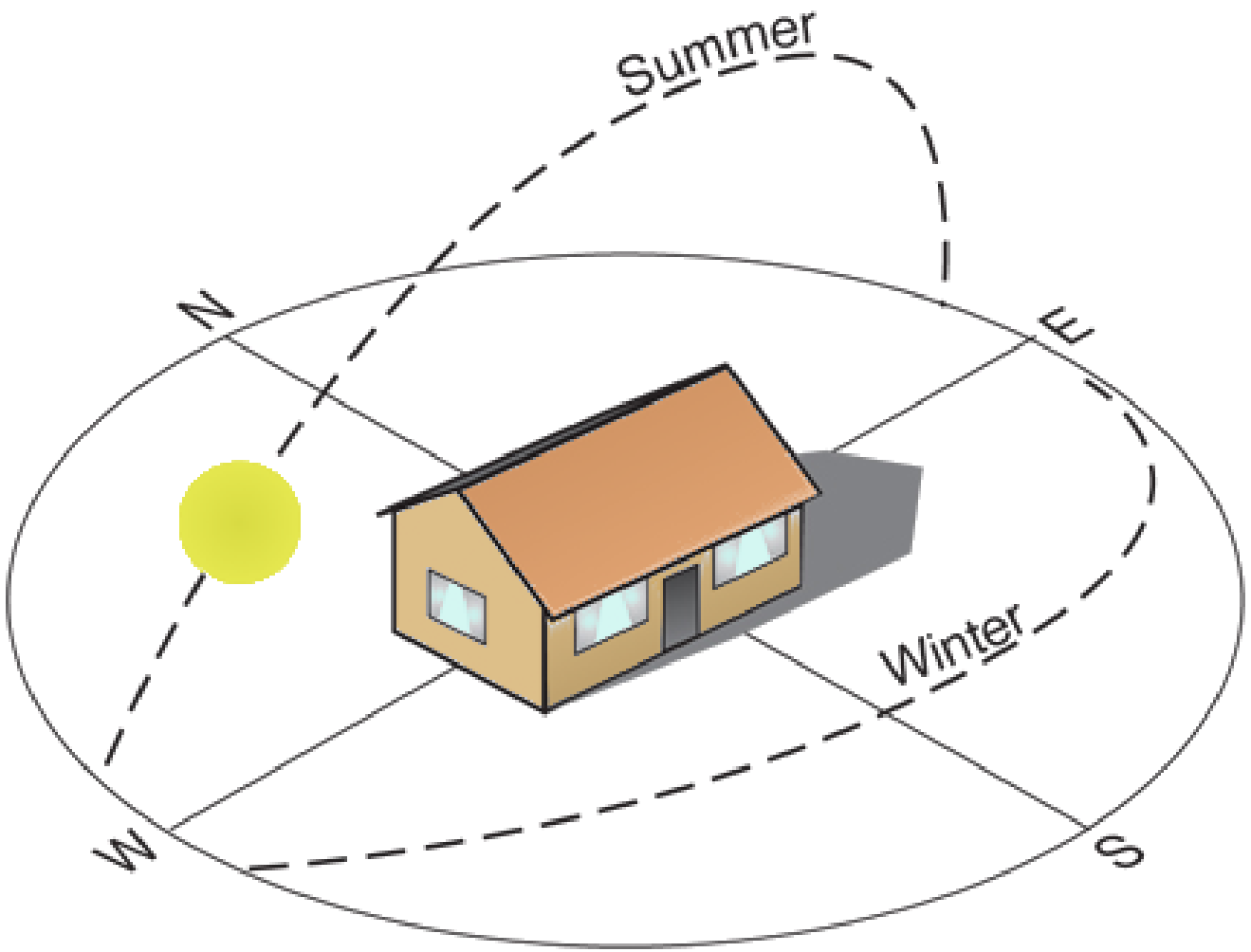
ECONOMY- A building should have min floor area with max utility.

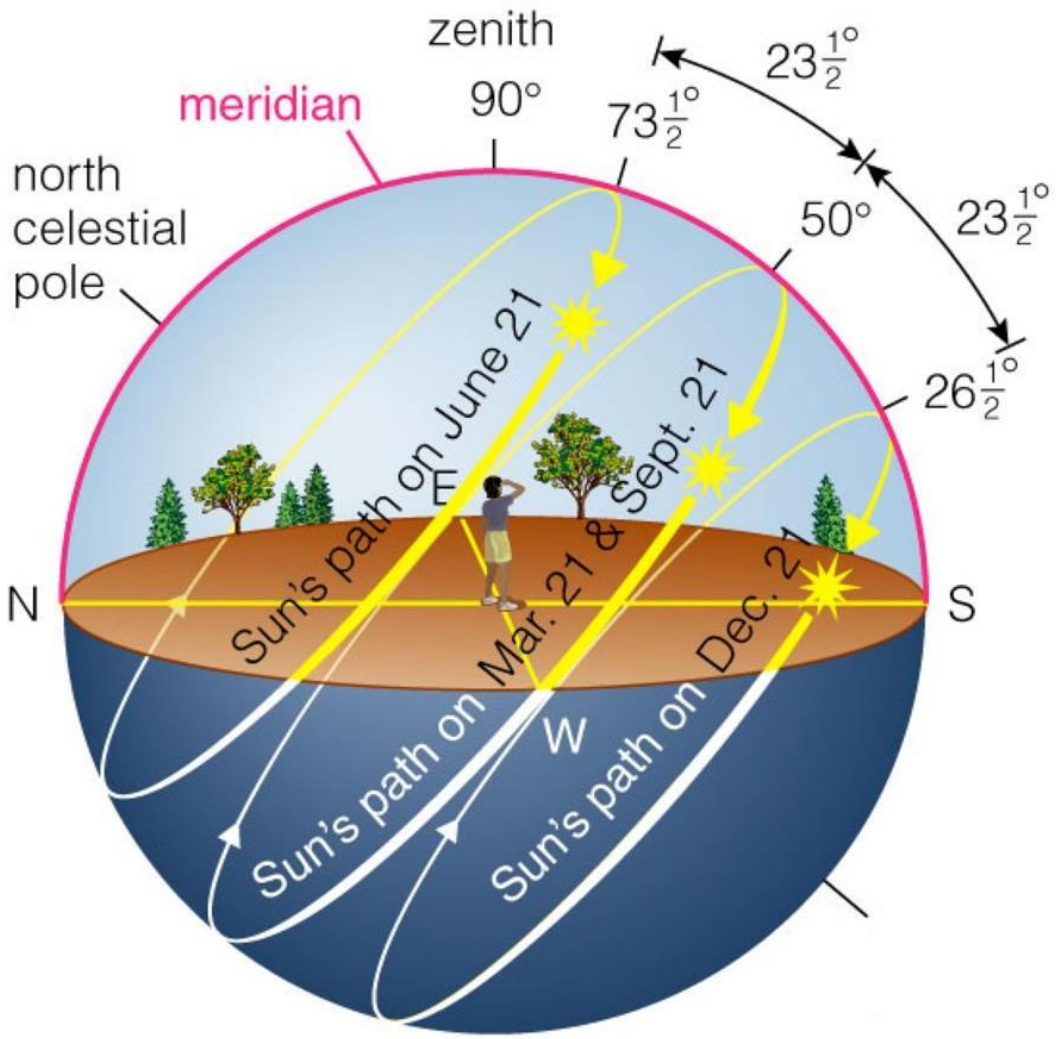
It should not achieved at the cost of strength. Only with proper planning and utility of space being maximized (passage being minimized)



Source- Houseplans

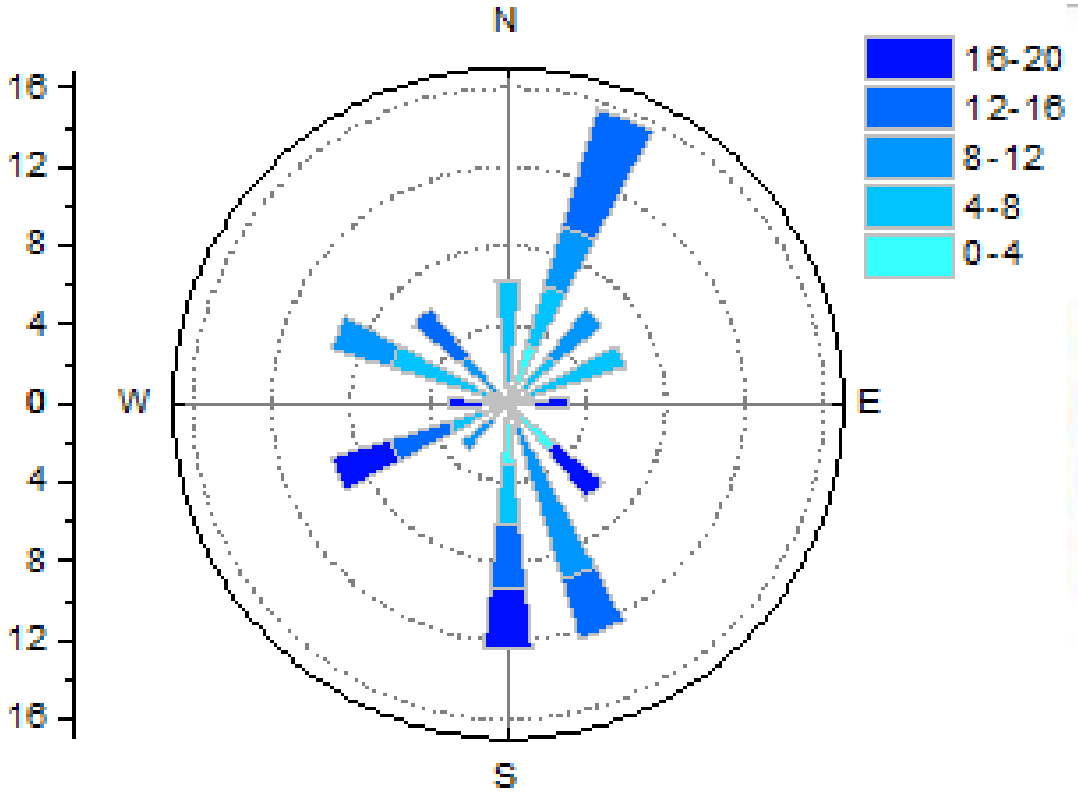
1.2 Significance of Sun Diagram

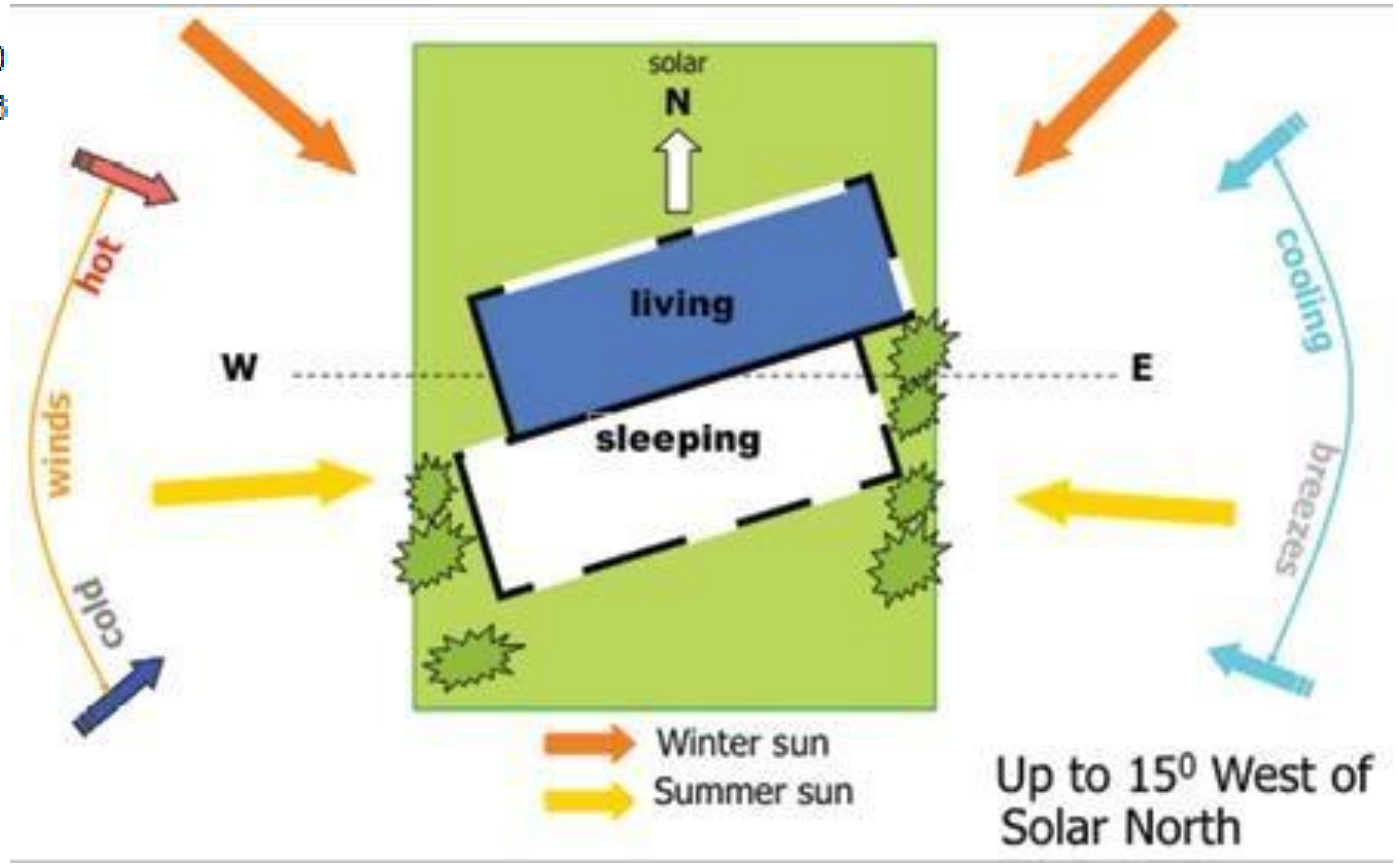




Source- GrahamHancock

1.3 Significance of Wind Diagram

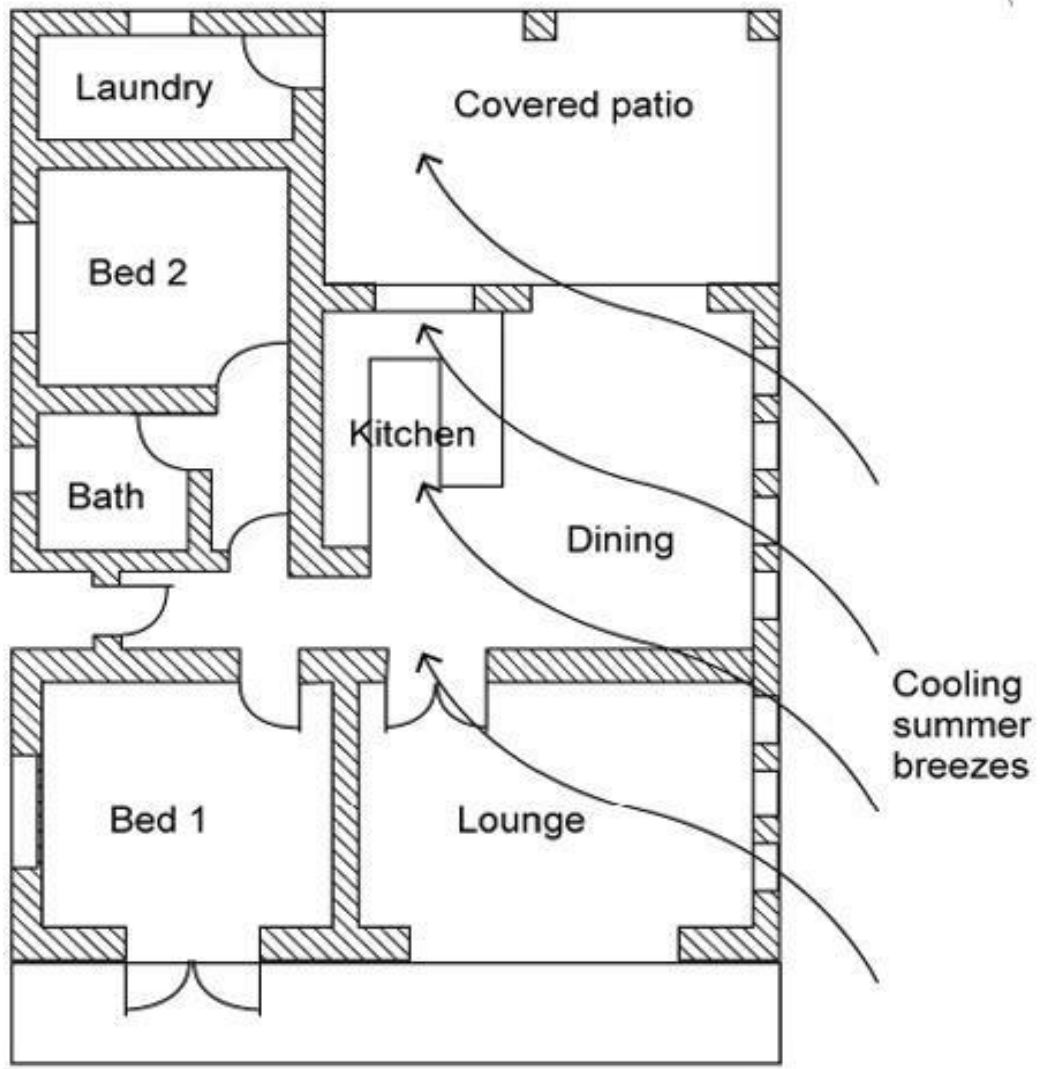




Source- Pinterest

1.4 Orientation

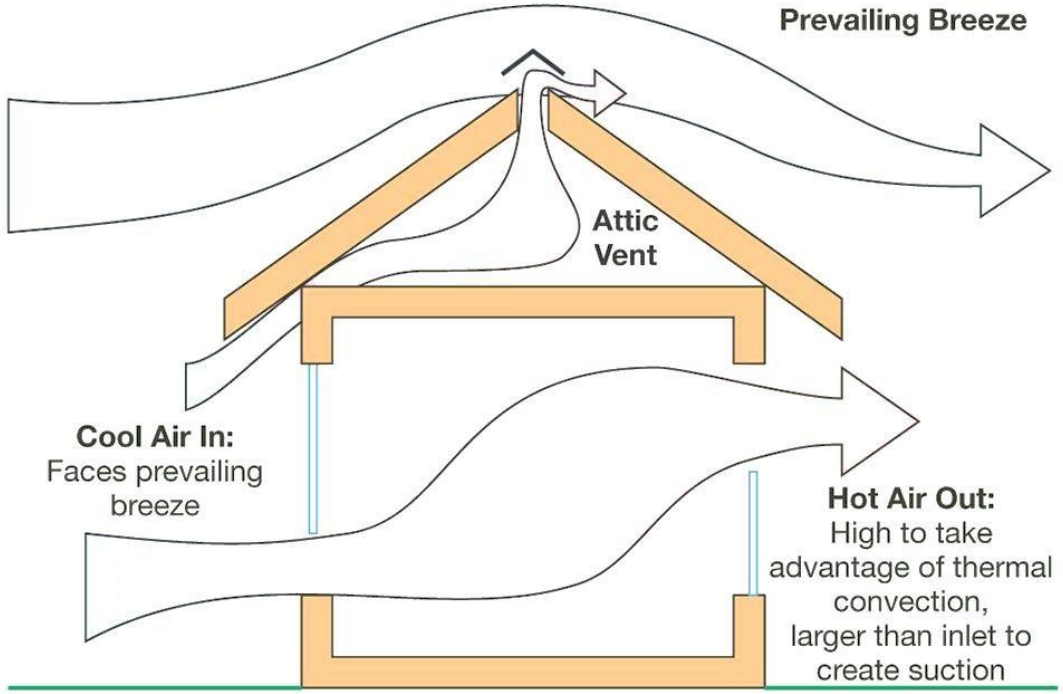
- Orientation is the positioning of a building in relation to seasonal variations in the sun's path as well as prevailing wind patterns.
- Good orientation can increase the energy efficiency of home, making it more comfortable to live in and cheaper to run.

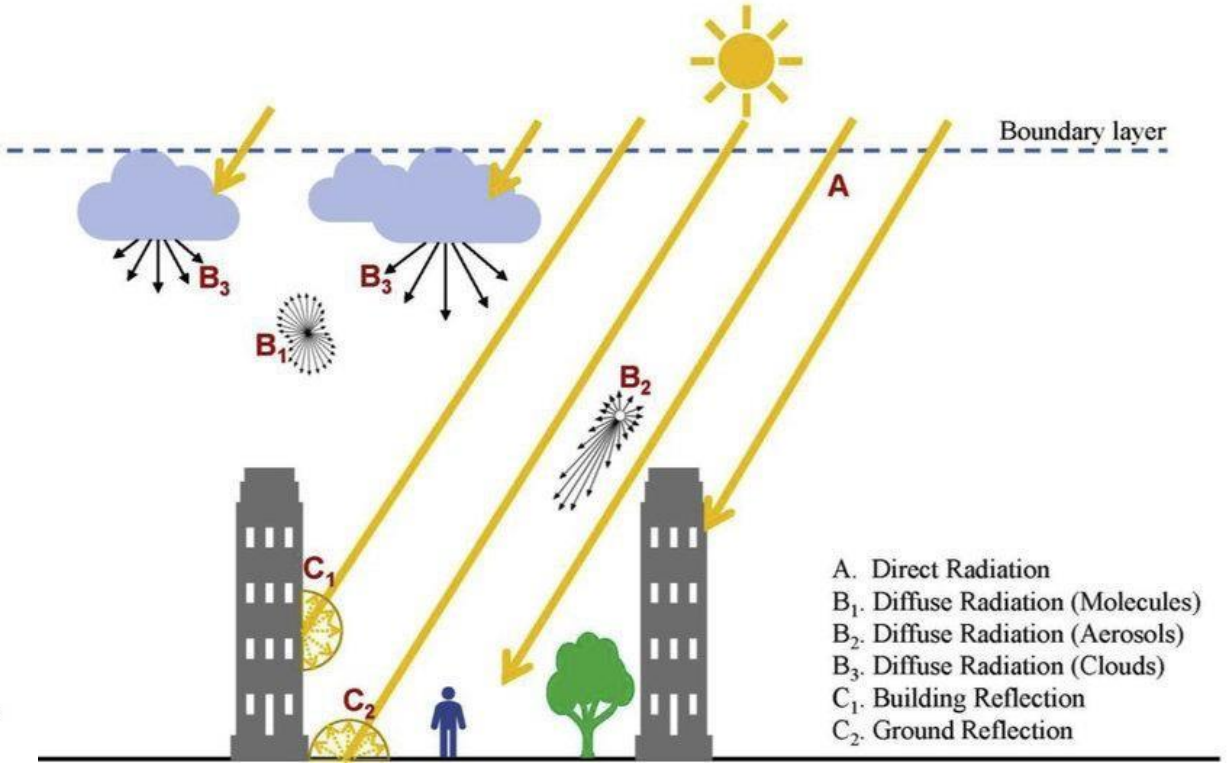


Source- Yourhome.gov.in

1.4.1 Orientation Factors Affecting,

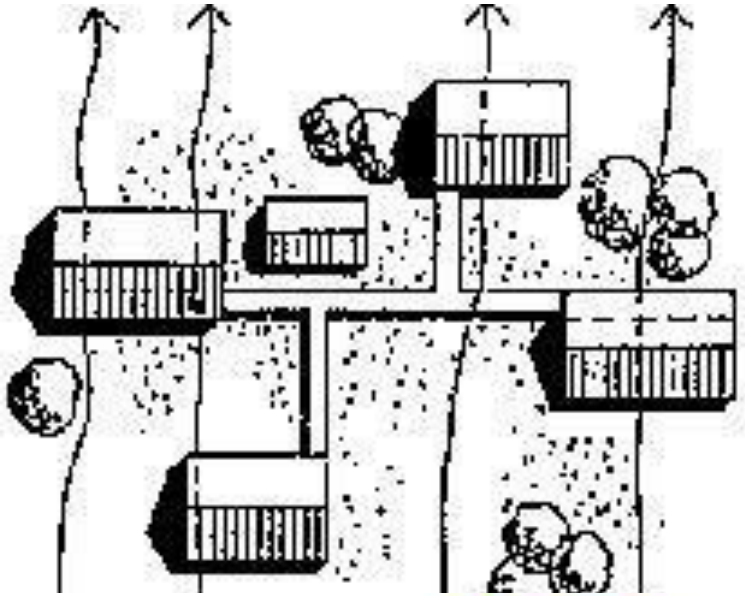
1. Solar radiation and temperature-
2. Clouds and Rains
3. Humidity
4. Humidity design consideration
5. Prevailing winds

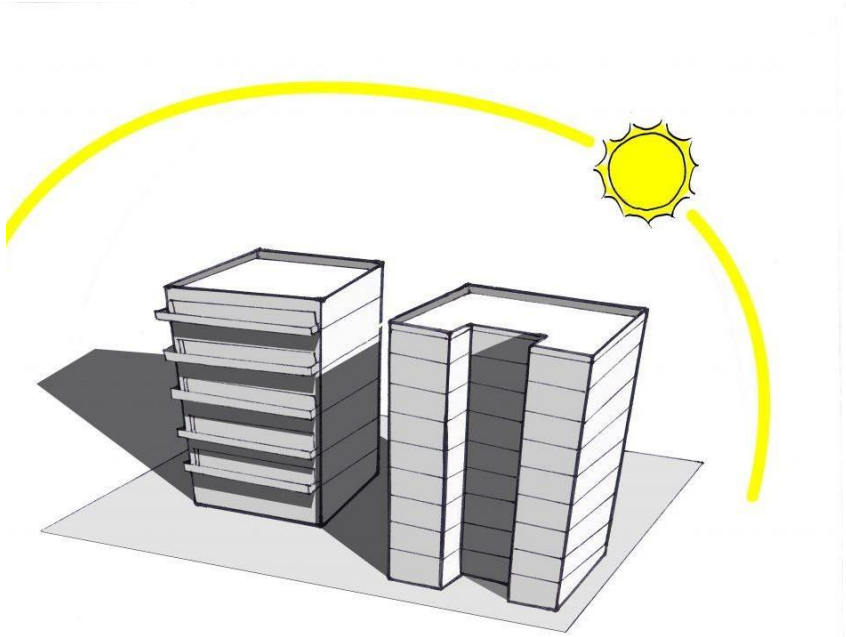


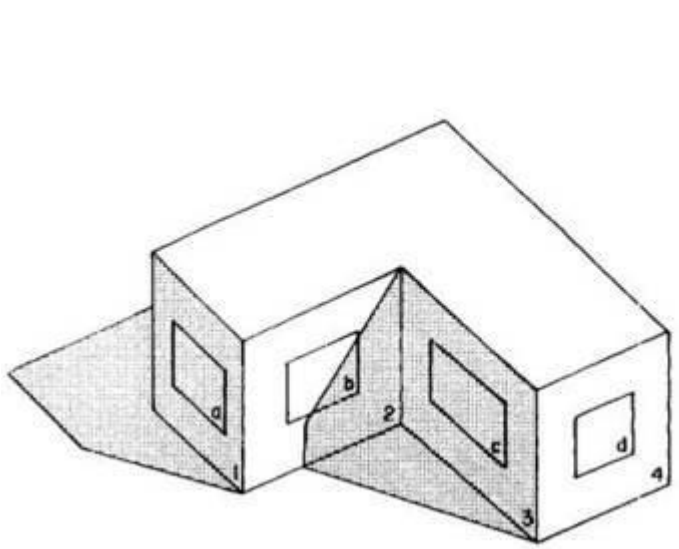


Source- Yourhome.gov.in

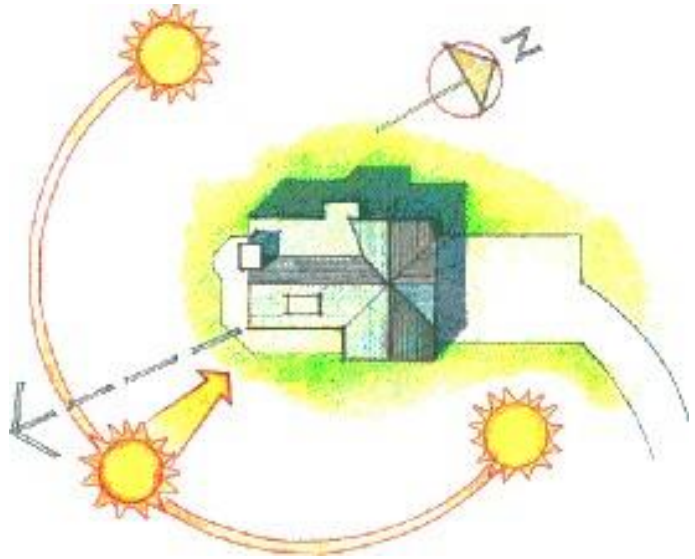
1.4.2 Orientation Criteria Under Indian Condition,











Source- Ecohomes



1.5 Concept of Green Building: Aspect at Planning Level, Construction Stage and Operational Level.

- A ‘green’ building is a building that, in its design, construction or operation, reduces or eliminates negative impacts, and can create positive impacts, on our climate and natural environment.
- Green buildings preserve precious natural resources and improve our quality of life.

Source- Pinterest

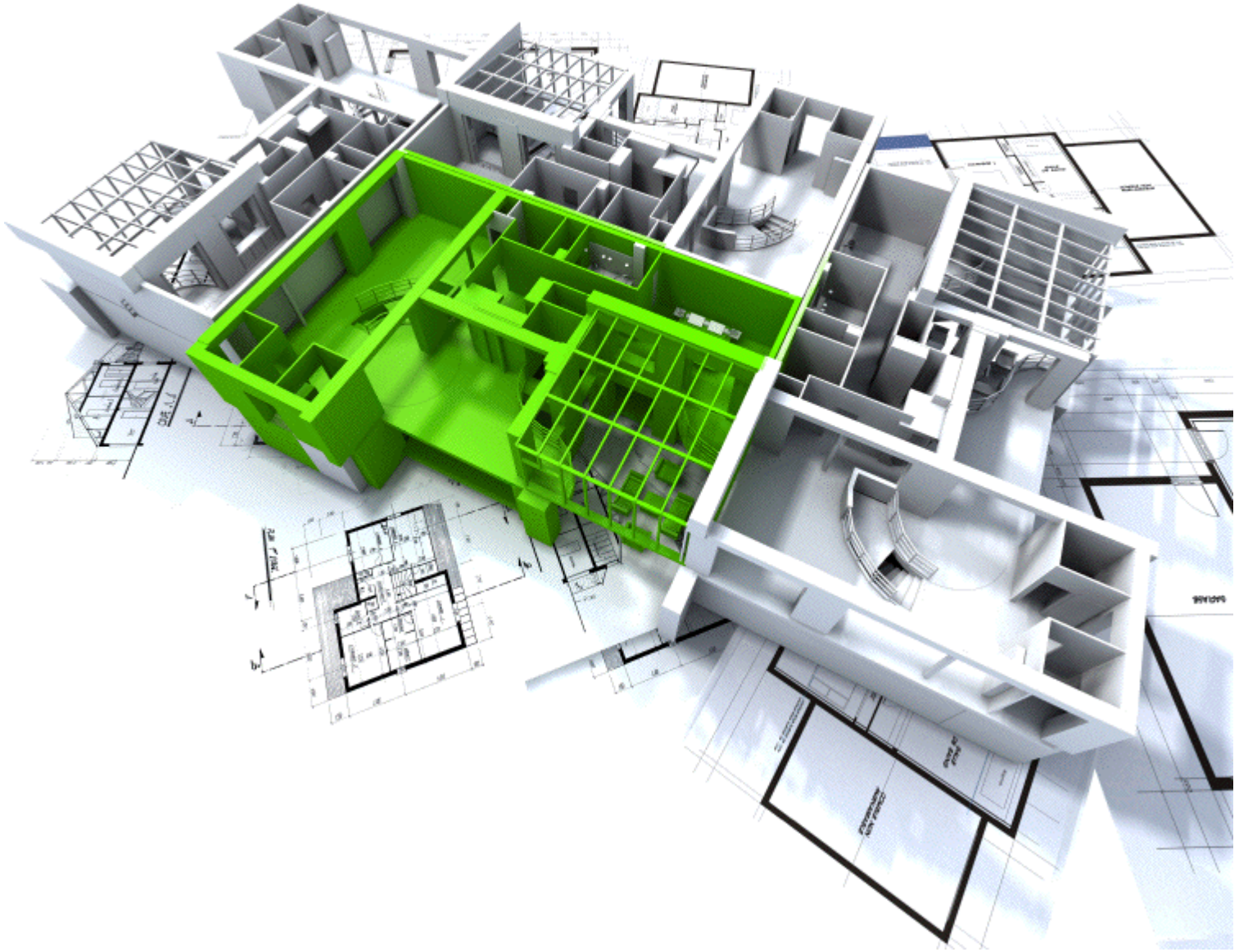
1.5.1 Aspect at Planning Level

- Efficient use of energy, water and other resources.
- Use of renewable energy, such as solar energy.
- Pollution and waste reduction measures, and the enabling of re-use and recycling.
- Good indoor environmental air quality.
- Use of materials that are non-toxic, ethical and sustainable.
- Consideration of the environment in design, construction and operation.
- Consideration of the quality of life of occupants in design, construction and operation.
- A design that enables adaptation to a changing environment.

Source- Pinterest

1.5.2 Construction Stage and Operational Level.

- Energy
- Indoor Air Quality
- Materials
- Site
- Waste
- Water
- Marketability
- Commissioning
- Community



Source- Greenspace

1.1 What is building bylaws?

- The rules and regulation framed by town planning authorities covering the requirements of building, ensuring safety of the public through open spaces, minimum size of rooms and height and area limitation, are known as building bye-laws.
- Rules and regulations which largely regulate the building activity should be formulated to get disciplined growth of building and the better planned development of towns and cities.

Principles underlying building bye-laws

- Classifying the building with unit as a family and mentioning the requirement.
- Classify rooms according use and then specifying minimum standard of each room with respect to size, height, floor area, ventilation and light.
- Specify height of compound wall and location of gate in wall.
- Controlling projection in marginal space.
- Insisting on suitable FSI or FAR.
- Specify suitable arrangements with respect to drainage and water supply.
- Specify set-backs, light plan and margin.
- Specify minimum size of plots, their dimension and figure.

Slum



Development **as per** laws



Source- Google

1.1.1 Objective of building bye-laws

- Pre-planning of building activity.
- Allow orderly growth and prevent haphazard development.
- Provisions of by-laws usually afford safety against fire, noise, health hazard and structure failure.
- Provide proper utilization of space to achieved maximum efficiency in planning.
- They provide health, safety and comfort to the people who live in building.
- Due to these bye-laws, each building will have proper approaches, light, air and ventilation.

1.1.2 Scope of building bye-laws as per NBC-SP7

Aspects of different type of building in building bye-laws:

1. Building frontage line
2. Minimum plot size
3. Built up area of building
4. Height of building
5. Provision of safety, water supply, drainage, proper light and ventilation
6. Requirement for off street parking space
7. Size of structural element

1.1.3 Applicability of building bye-laws group- 1 to 5

- New construction
- Additional and alternations to a building
- Changing of occupancy of building (residential → education etc.)
- Development of land is undertaken
- In demolition

1.1.4 Different types of area

- Built up area
- Plinth area
- Floor area
- Carpet area
- Super built up area

1.1.4.1 Built up area

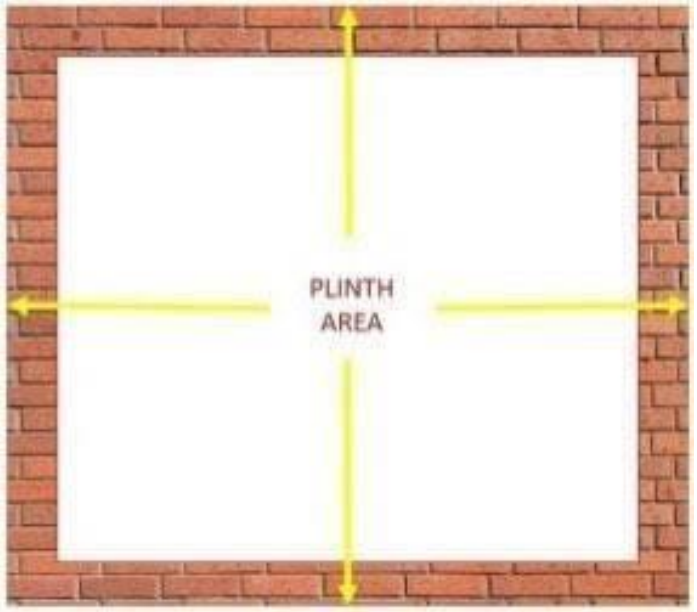
- “The area covered by the building immediately above the plinth level”
- It is also known as covered area.

Details to be included	Details not to be included
Basement	Steps
Building Structure	Otta
Ramps	Kundi
Swimming pool	Soak pit
Reservoir	Fountain
Chimney	Manhole
Water purification plant	Water tank

1.1.4.2 Plinth area

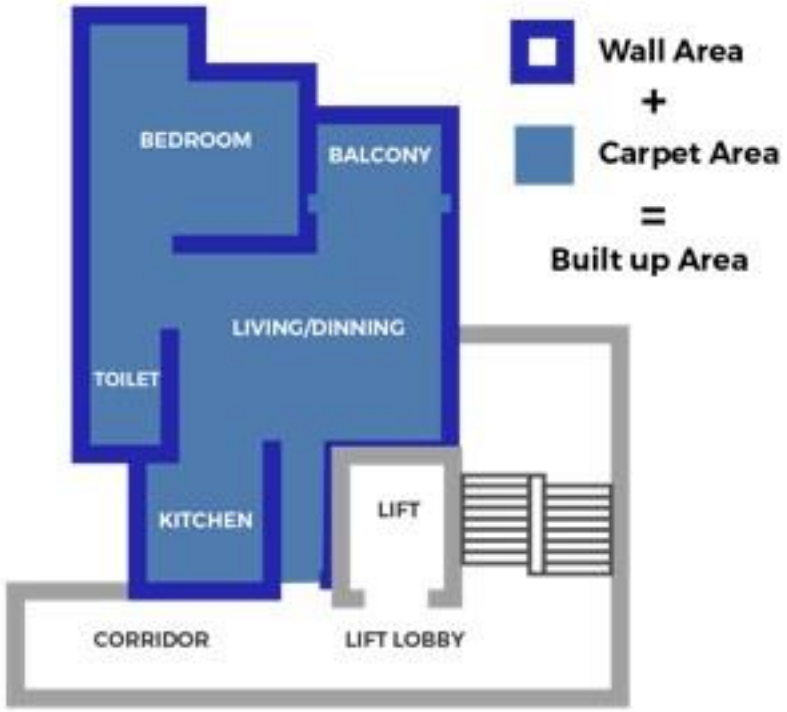
Means the built-up covered area measured at the floor level of the basement or of any storey.

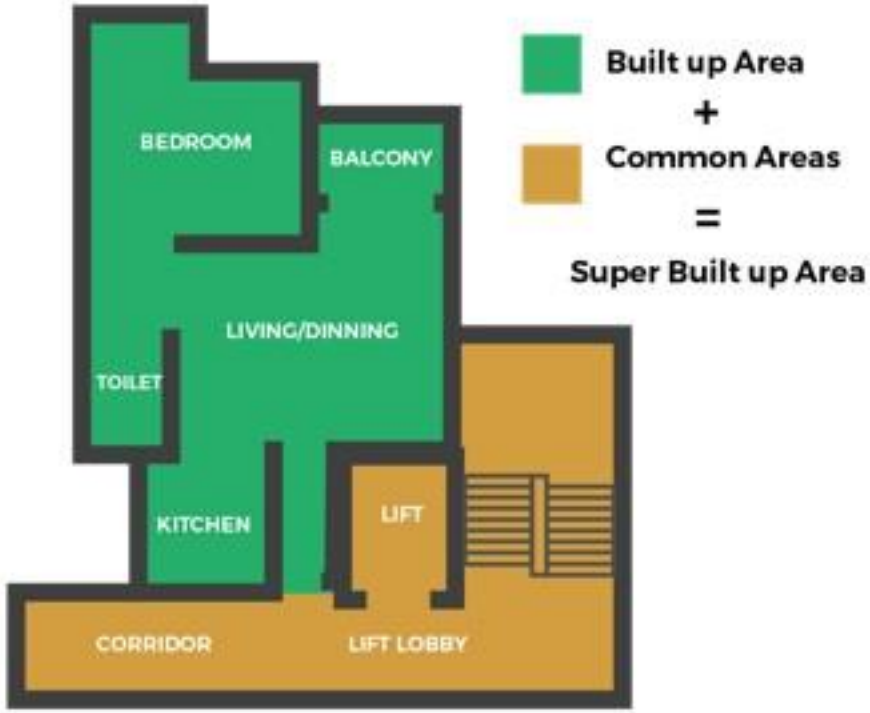
Area to be included in plinth area	Area not to be included in plinth area
Area of walls at floor level excluding plinth offset, external area of projecting cladding.	Additional floor to accommodate seats in theatre, assembly hall, or auditorium
Cover of stair	Balcony
Barsati	Curvature provided to improve appearance
Machine room	Platform without cover terrace above floor
Porch without cantilever	Dome, tower at terrace, spiral stairs



Source- Housemap







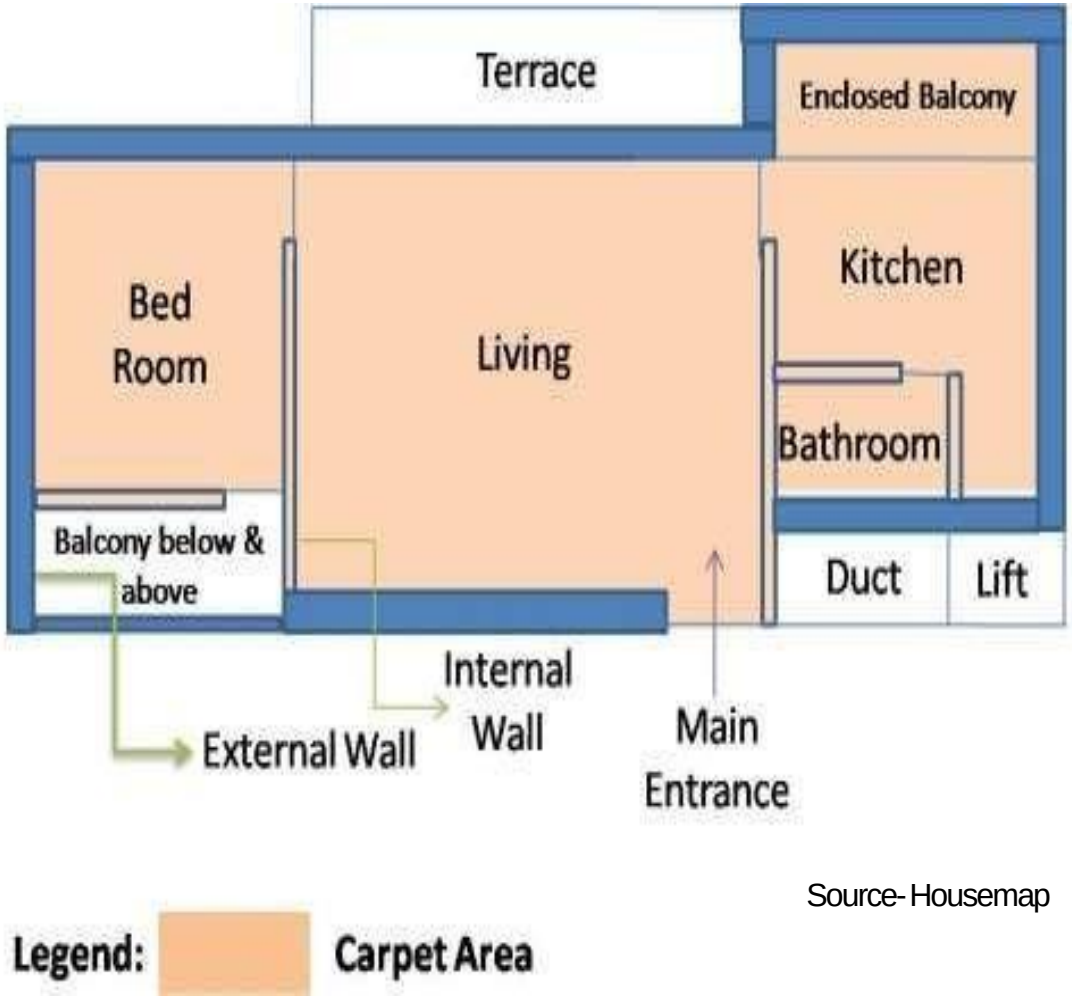
CARPET AREA **BUILT UP AREA** **SUPER BUILT UP AREA**

Difference between Carpet area, Built-up area and Super Built up Area

Source- Housemap

1.1.4.3 Carpet area

- “it is net usable area within building”.
- Means the net area within a unit of a building excluding walls or columns, service areas such as washrooms, kitchen and pantry, semi- open spaces such as veranda or balcony, and corridors or passages.

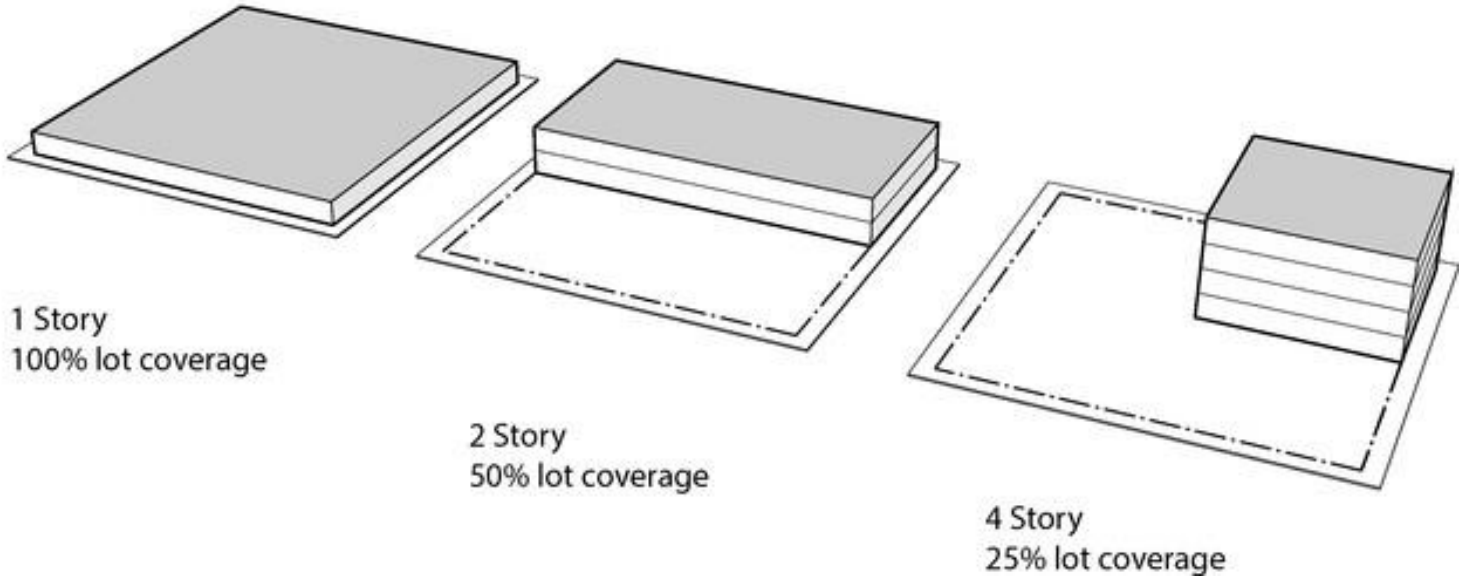


Source- Housemap

1.1.4.4 Floor area

Floor Area Ratio (FAR)
1:1 Ratio

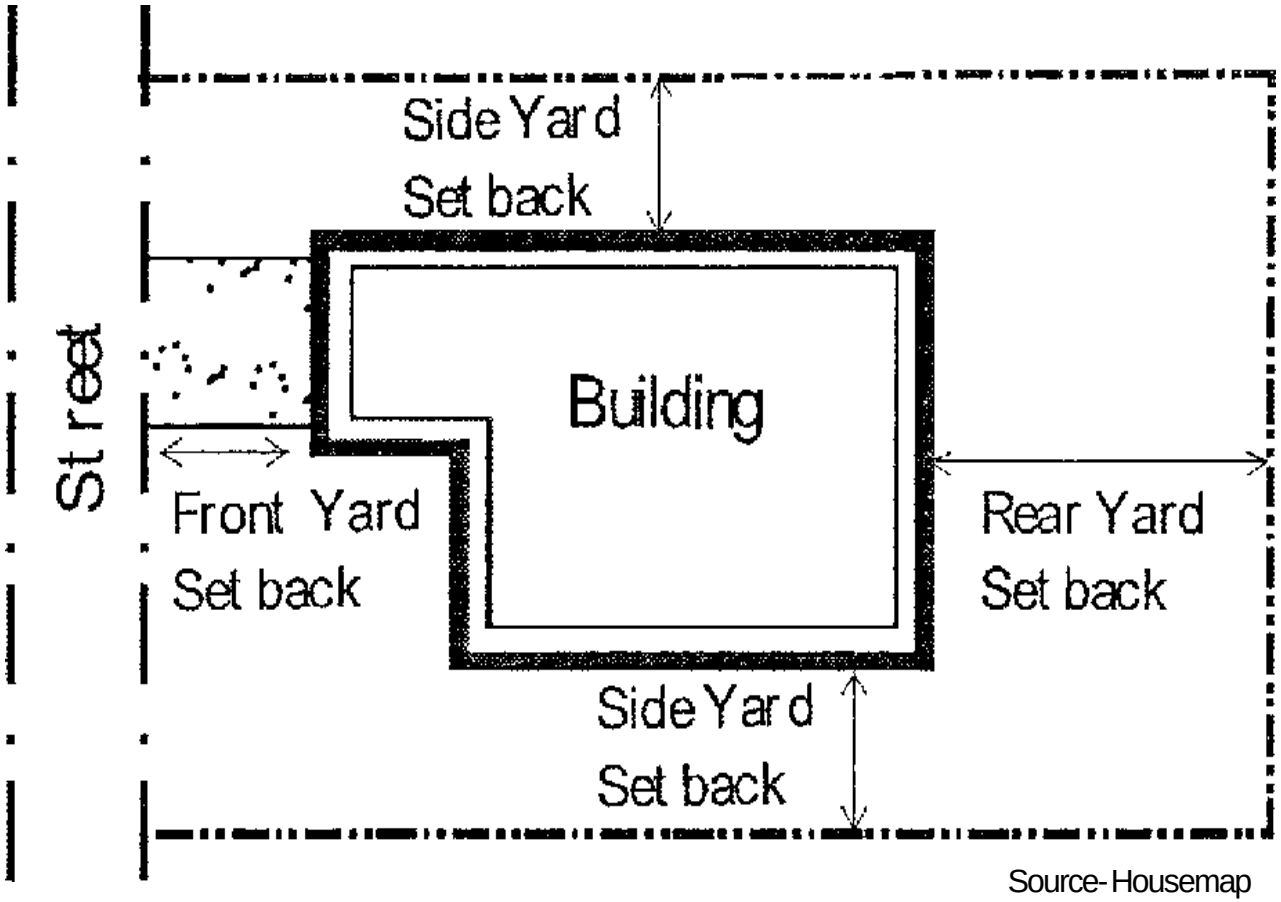
- “It is usable area on each floor”
- It including,
  - Porch
  - Each flat area in single floor for apartment
  - Stairs
  - Corridor
  - door swing place



Source- Housemap

1.1.4.5 Set back distance

- Better condition of air, light and ventilation.
- At corners it improve visibility and safety from traffic.
- Space for parking.
- Protection of building from street nuisance like noise.
- Reduce the danger of fire from neighbor house.
- It provide privacy in building.

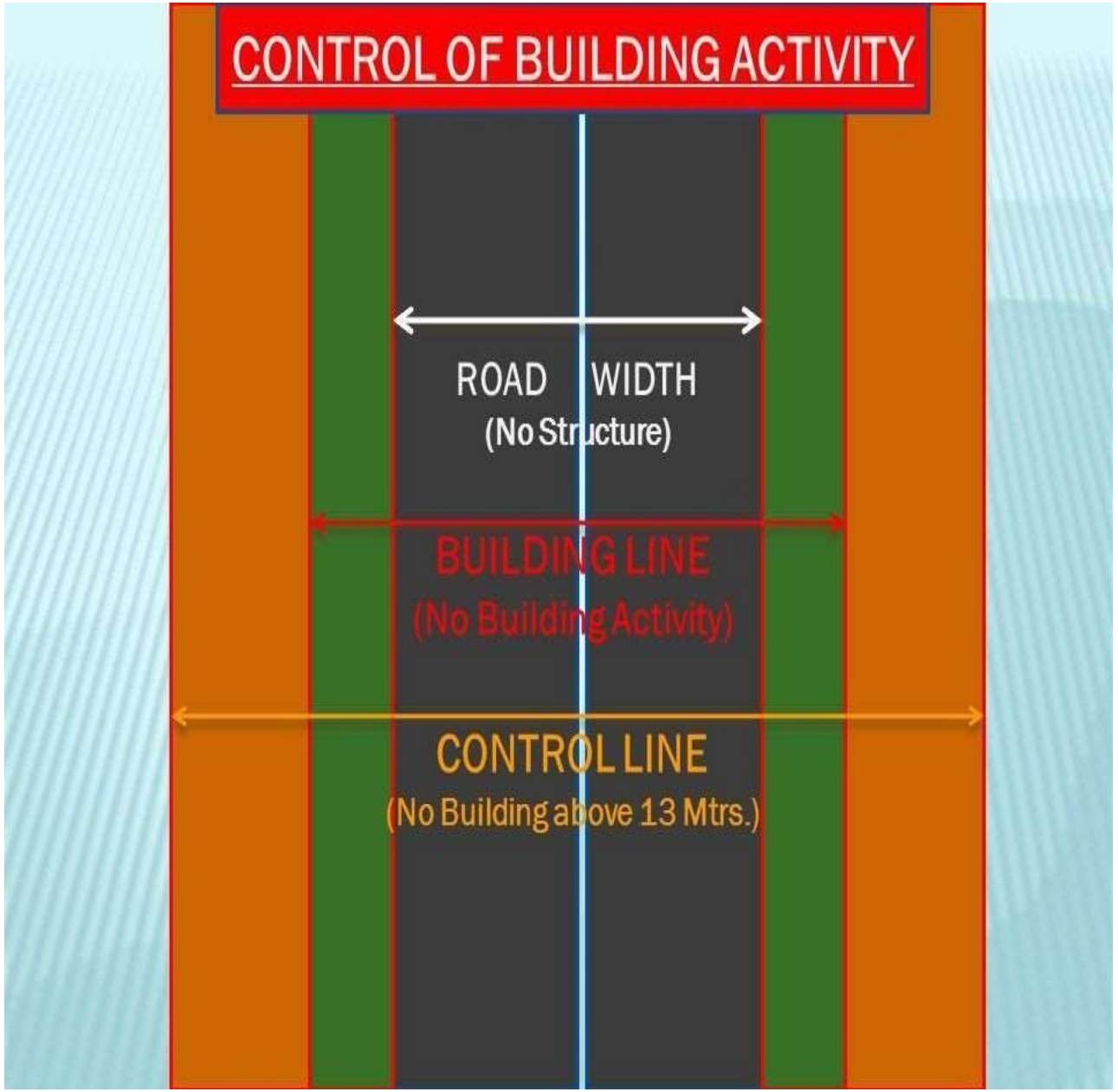


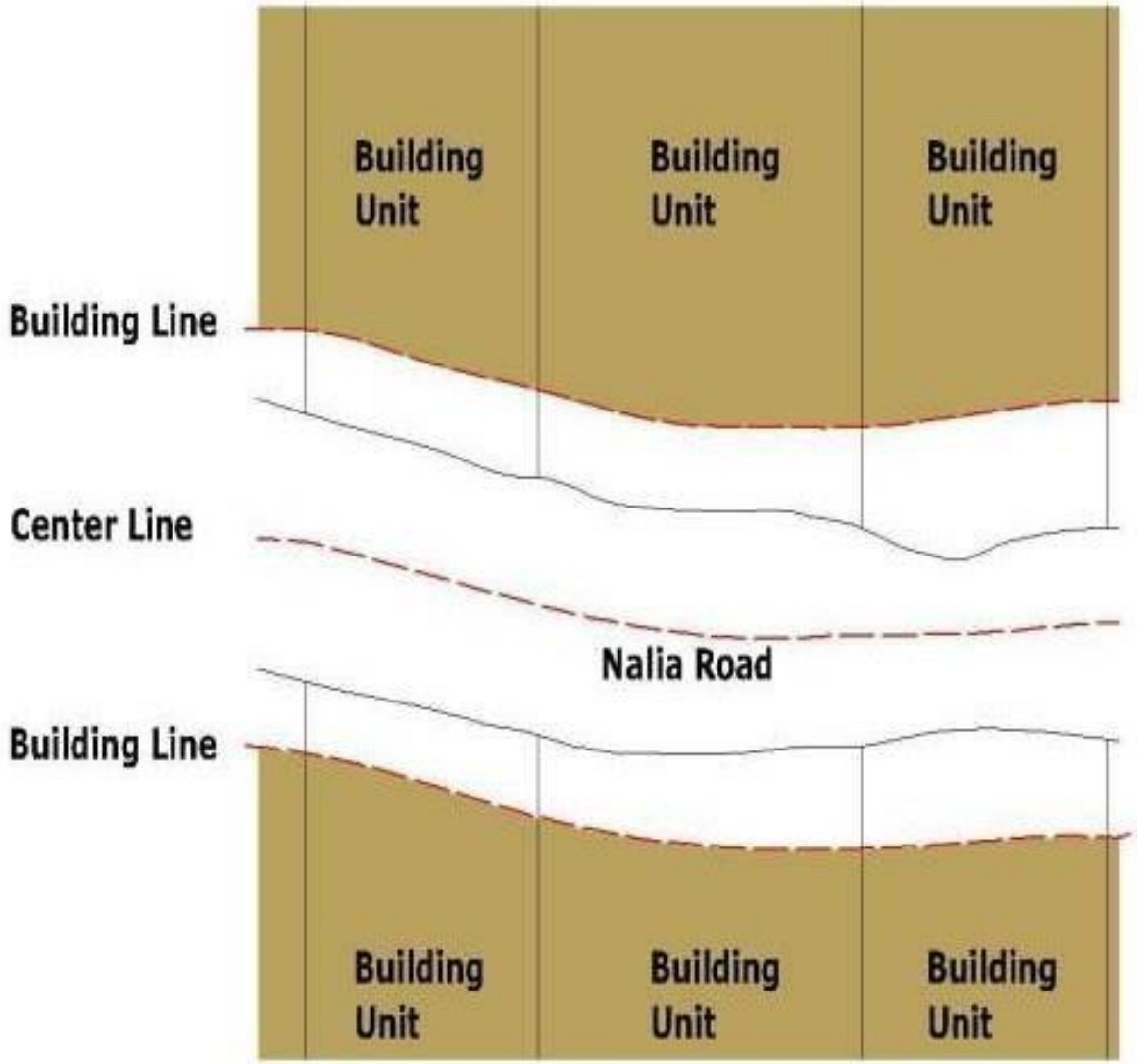
Source- Housemap

Width of road	Permissible set back
Up to 12 m	1.5
>12 m	2.0
Road width <3 m and length < 30 m	No set- back

1.1.4.6 Building & control line

- The line up to which the plinth of the building adjoining a street or road or on future street may lawfully extend is know as building line.
- Building such as cinemas, factories, commercial center which attract large number of vehicle should be set-back distance apart from the building line.
- The line up to which such building can be constructed is known as control line.



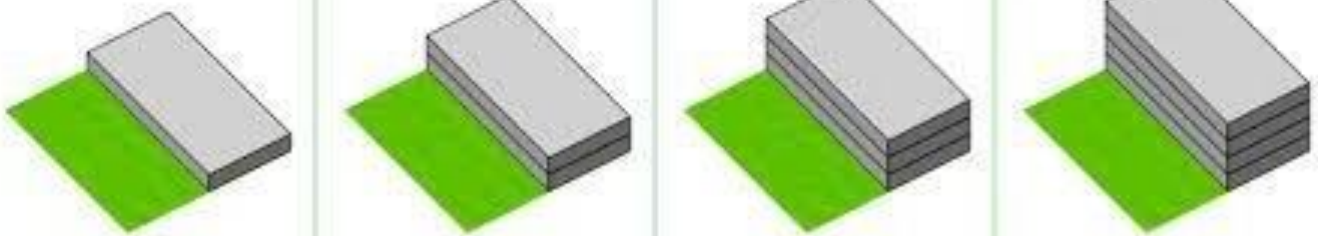
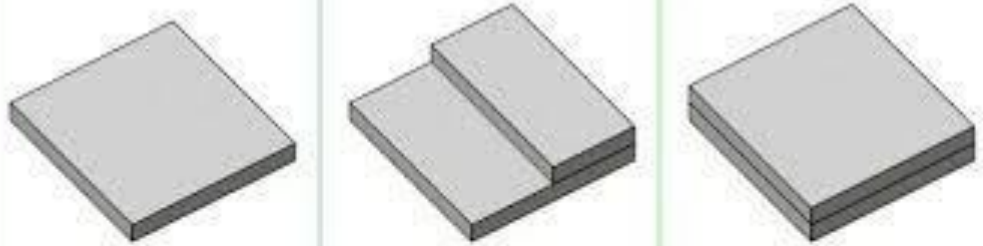


Source- controllines

1.1.5 Floor space index

$$FSI = \frac{Total\ covered\ area\ of\ all\ floors}{plot\ area}$$

Zone	Permissible FSI	Remark
Residential(scheme area)	1	Max at G.F 0.4
Residential(city area)	4	Width of road more than 12m
Residential(city area)	3	Width of road less than 12m
Industrial (scheme area)	1.5	Maximum allowed on G.F 1.0
Commercial (scheme area)	1.33	Maximum allowed on G.F 0.4

FAR / FSI / BCR	0.25 25%	0.5 50%	1 100%	1.5 150%	2 200%
25%					
50%	not possible				
100%	not possible	not possible			

Source- Housemap

Building Bye-laws For Residential Area

- Size of plots
- Area limitation
- Margin
- Height of the building
- Plinth height
- Minimum area of rooms
- Basement
- Compound wall
- Projection margin
- Parapet wall
- Mezzanine floor
- staircase

No.	Name of room	Minimum area	Minimum size of side	Height	Other requirement
1.	Habitable room -bed room -living room -study room	9.5 m^2	2.4 m	Not less then 2.75 m from surface of floor to lowest point of ceiling	As per need
2.	Kitchen	5 m^2	1.8 m	Not less than 2.75 m	As per need
3.	Bath rooms and water closets	1.8 m^2	If bath and water closet are combined, its floor area shall not less than 2.8 m^2 with minimum width 1.2 m	Not less than 2.1 m	As per need
4.	Store room	3 m^2	-----	Not less than 2.2 m	-----
5.	Garage	12.5 m^2	3m x 6m	Not less than 2.4 m	-----
6.	Stair case		The minimum width of stair is 0.9 m.	Clear head room shall be 2.2 m	• Minimum width of tread without nosing 250 mm in residential. • The maximum height of rise shall be 190 mm for residential building.

23.8.7 BUILDING REQUIREMENTS

SR.NO.	UNIT	MIN FLOOR AREA(SQ.MT)	Min Side (mt)
a.	Room, shop, office	9.00	2.40
b.	Kitchen	3.80	1.80
c.	Store room	3.60	1.80
d.	Puja, bath, pump, coal & dressing rooms	1.35	0.90
e.	Garage	12.0	2.7wx2.0h
f.	W.c, wash	0.81	0.90
g.	Study room	3.60	1.80
h.	stall	up to 9.0	1.80
i.	toilet	1.75	1.00
LOFT NOT EXCEEDING 1/3rd OF THE AREA OF THE ROOM SHALL BE PERMITTED AT A HEIGHT OF 2.0MTRS.			

Source- Code-7.3.6

Planning of residential building: bungalows,

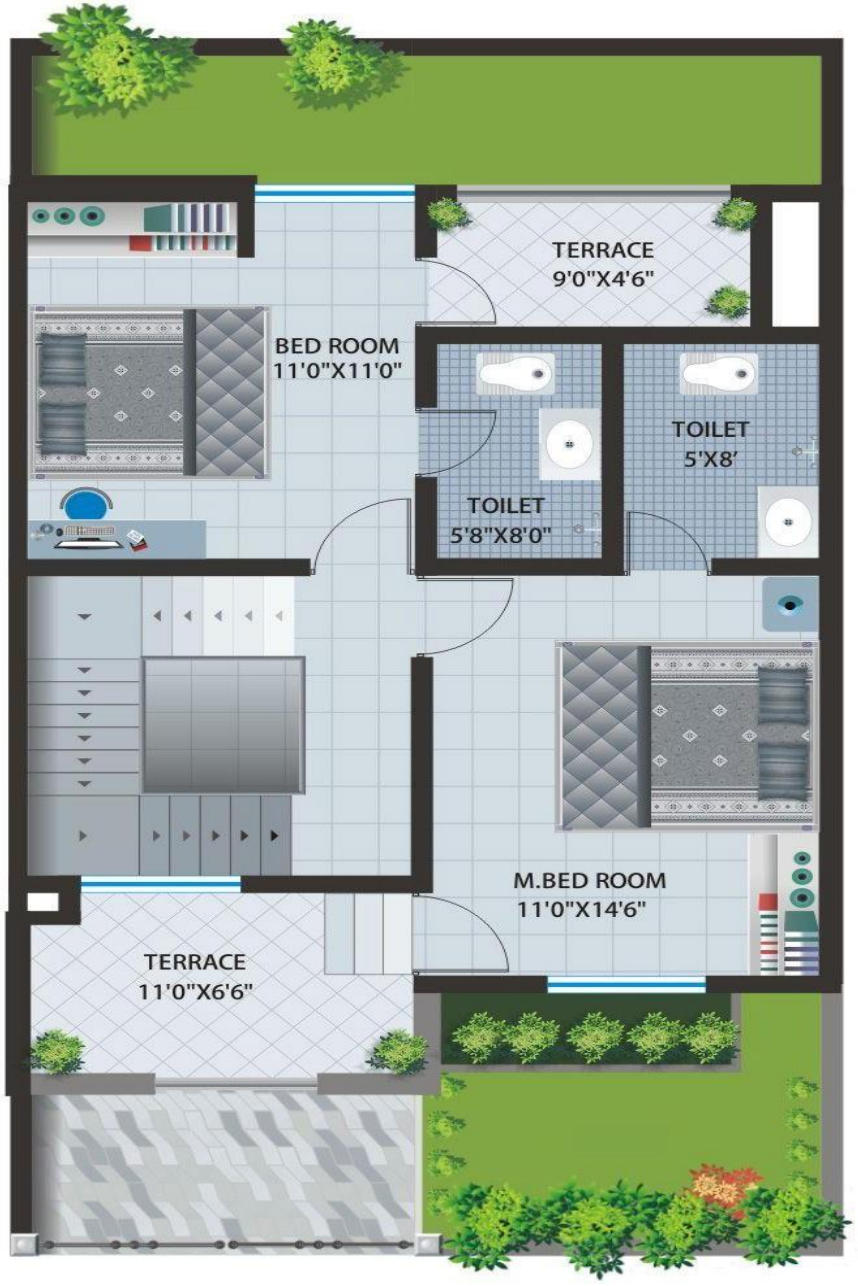


Source- Indiamart Services

Planning of residential building: row bungalows,



GROUND FLOOR PLAN



FIRST FLOOR PLAN

Source- Indiamart Services

Planning of residential building: apartments,



Source- Indiamart Services

Planning of residential building: twin bungalows,



GROUND FLOOR PLAN



FIRST FLOOR PLAN

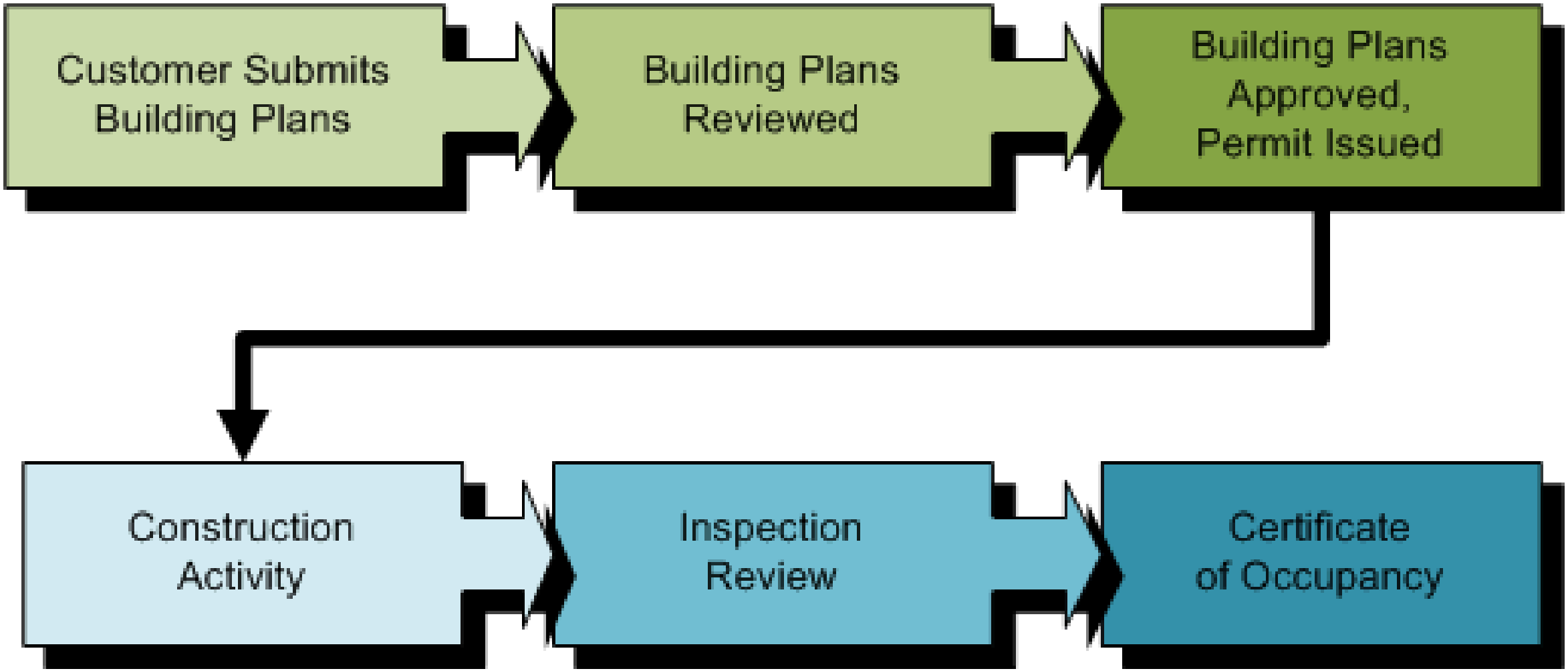
Source- Indiamart Services

PLOT AREA : 1250 SQ.FT.

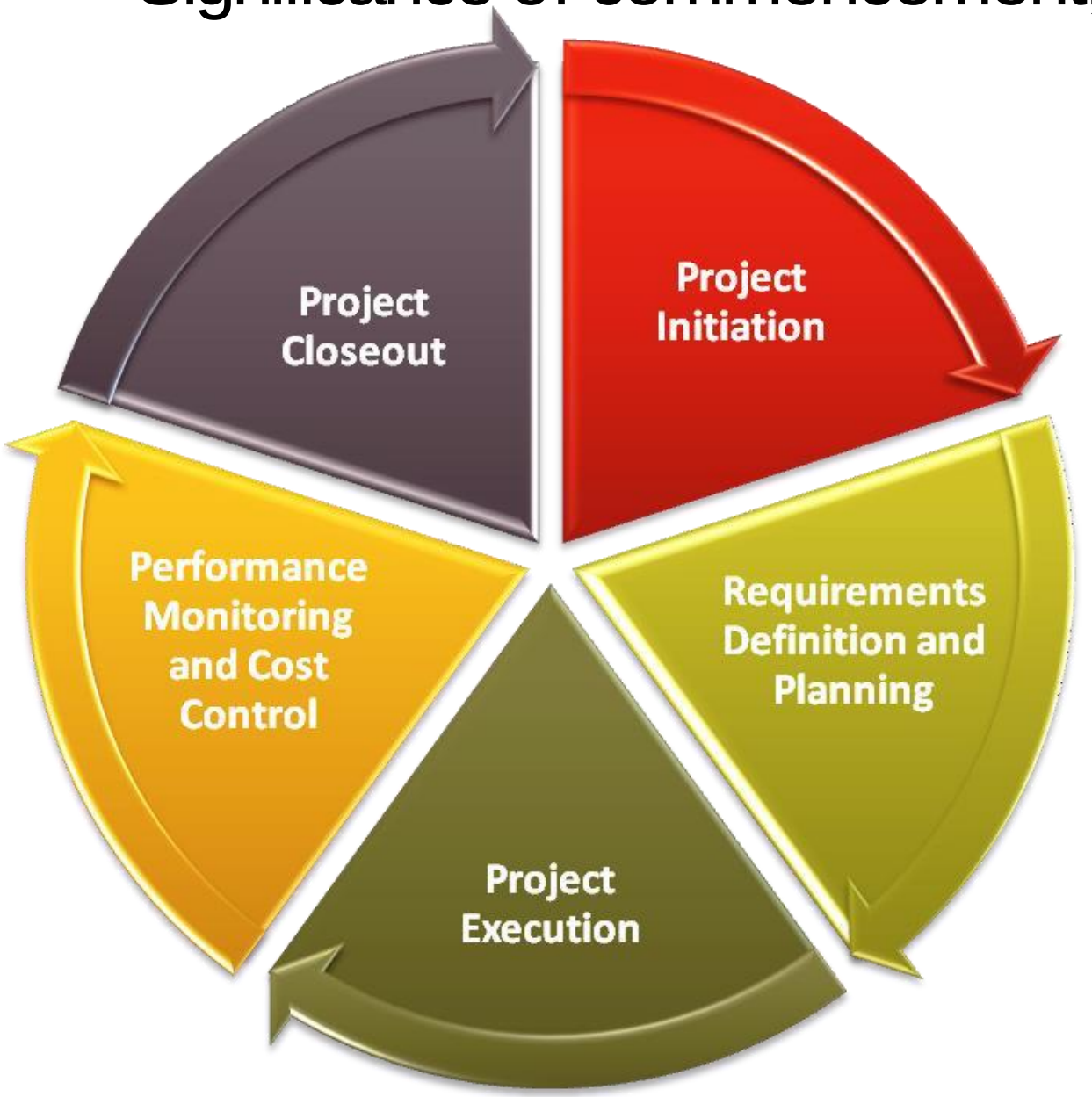
Planning of residential building: twin bungalows,



Procedure of building permission,



Significance of commencement,





These are the typical stages in any construction project.
Good planning at the beginning and anticipation of contingencies will help the entire process go much more smoothly.

Plinth completion or occupancy certificate

